

**THE FORTIETH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

December 1, 1979

Problem A-1

Find positive integers n and $a_1, a_2, \dots, a_n$ such that

$$a_1 + a_2 + \dots + a_n = 1979$$

and the product $a_1 a_2 \dots a_n$ is as large as possible.

Problem A-2

Establish necessary and sufficient conditions on the constant k for the existence of a continuous real valued function $f(x)$ satisfying $f(f(x)) = kx^9$ for all real x .

Problem A-3

Let $x_1, x_2, x_3, \dots$ be a sequence of nonzero real numbers satisfying

$$x_n = \frac{x_{n-2}x_{n-1}}{2x_{n-2} - x_{n-1}} \text{ for } n = 3, 4, 5, \dots$$

Establish necessary and sufficient conditions on x_1 and x_2 for x_n to be an integer for infinitely many values of n .

Problem A-4

Let A be a set of $2n$ points in the plane, no three of which are collinear. Suppose that n of them are colored red and the remaining n blue. Prove or disprove: there are n closed straight line segments, no two with a point in common, such that the endpoints of each segment are points of A having different colors.

Problem A-5

Denote by $[x]$ the greatest integer less than or equal to x and by $S(x)$ the sequence $[x], [2x], [3x], \dots$. Prove that there are distinct real solutions α and β of the equation $x^3 - 10x^2 + 29x - 25 = 0$ such that infinitely many positive integers appear both in $S(\alpha)$ and in $S(\beta)$.

Problem A-6

Let $0 < p_i < 1$ for $i = 1, 2, \dots, n$. Show that

$$\sum_{i=1}^n \frac{1}{|x - p_i|} < 8n \left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1} \right)$$

for some x satisfying $0 < x < 1$.

Problem B-1

Prove or disprove: there is at least one straight line normal to the graph of $y = \cosh x$ at a point $(a, \cosh a)$ and also normal to the graph of $y = \sinh x$ at a point $(c, \sinh c)$.

[At a point on a graph, the normal line is the perpendicular to the tangent at that point. Also, $\cosh x = (e^x + e^{-x})/2$ and $\sinh x = (e^x - e^{-x})/2$.]

Problem B-2

Let $0 < a < b$. Evaluate

$$\lim_{t \rightarrow 0} \left\{ \int_0^1 [bx + a(1-x)]^t dx \right\}^{1/t}$$

[The final answer should not involve any operations other than addition, subtraction, multiplication, division, and exponentiation.]

Problem B-3

Let F be a finite field having an odd number m of elements. Let $p(x)$ be an irreducible (i.e., nonfactorable) polynomial over F of the form

$$x^2 + bx + c, \quad b, c \in F.$$

For how many elements k in F is $p(x) + k$ irreducible over F ?

Problem B-4

(a) Find a solution that is not identically zero, of the homogeneous linear differential equation

$$(3x^2 + x - 1)y'' - (9x^2 + 9x - 2)y' + (18x + 3)y = 0.$$

Intelligent guessing of the form of a solution may be helpful.

(b) Let $y = f(x)$ be the solution of the *nonhomogeneous* differential equation

$$(3x^2 + x - 1)y'' - (9x^2 + 9x - 2)y' + (18x + 3)y = 6(6x + 1)$$

that has $f(0) = 1$ and $(f(-1) - 2)(f(1) - 6) = 1$. Find integers a, b, c such that $(f(-2) - a)(f(2) - b) = c$.

Problem B-5

In the plane, let C be a closed convex set that contains $(0,0)$ but no other point with integer coordinates. Suppose that $A(C)$, the area of C , is equally distributed among the four quadrants. Prove that $A(C) < 4$.

Problem B-6

For $k = 1, 2, \dots, n$ let $z_k = x_k + iy_k$, where the x_k and y_k are real and $i = \sqrt{-1}$. Let r be the absolute value of the real part of

$$\pm \sqrt{z_1^2 + z_2^2 + \dots + z_n^2}.$$

Prove that $r < |x_1| + |x_2| + \dots + |x_n|$.